Geometry & Coordinate Geometry

ANSWER KEY

Section 1: Vectors & Dot Products

1 What is the dot product of the vectors $u = \langle 3, -4 \rangle$ and $v = \langle 2, 5 \rangle$?

a. -14 ✓

b. 26

c. 14

d. -26

$u \cdot v = (3)(2) + (-4)(5) = 6 - 20 = -14$.

2 For what value of k are the vectors $u = \langle k, 3 \rangle$ and $v = \langle 6, -2 \rangle$ perpendicular?

a. 1 ✓

b. 2

c. -1

d. 3

Perpendicular vectors have a dot product of zero: $6k + 3(-2) = 0 \Rightarrow 6k - 6 = 0 \Rightarrow k = 1$.

3 What is the magnitude (length) of the vector $v = \langle 5, -12 \rangle$?

a. 13 ✓

b. 17

c. 7

d. 169

$\|v\| = \sqrt{5^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

Section 2: Distance, Midpoint & Lines

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4 What is the distance between the points $A(1, 2)$ and $B(4, 6)$?

a. 5 ✓

b. 7

c. $\sqrt{7}$

d. 25

$d = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25} = 5.$

5 What are the coordinates of the midpoint of the segment joining $P(-2, 7)$ and $Q(6, 1)$?

a. $(2, 4)$ ✓

b. $(4, 8)$

c. $(2, 3)$

d. $(1, 4)$

Midpoint $M = \left(\frac{-2+6}{2}, \frac{7+1}{2}\right) = (2, 4).$

6 What is the slope of a line perpendicular to the line $3x - 4y = 8$?

a. $-\frac{4}{3}$ ✓

b. $\frac{3}{4}$

c. $\frac{4}{3}$

d. $-\frac{3}{4}$

Slope of $3x - 4y = 8$ is $m_1 = \frac{3}{4}$. The perpendicular slope is the negative reciprocal: $m_2 = -\frac{4}{3}$.

Section 3: Conic Sections (Parabola & Circle)

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7 What is the center and radius of the circle $(x-3)^2+(y+2)^2=36$?

a. **Center** $(3,\ -2)$**, radius** 6 ✓

b. Center $(-3, 2)$, radius 6

c. Center $(3,\ -2)$, radius 36

d. Center $(-3, 2)$, radius 36

In standard form $(x-h)^2+(y-k)^2=r^2$, the center is $(h, k)=(3,\ -2)$ and $r=\sqrt{36}=6$.

8 In the parabola $(y-1)^2=8(x+2)$, which direction does it open?

a. **To the right** ✓

b. To the left

c. Upward

d. Downward

Since y is squared and $4p=8>0$, the parabola opens horizontally to the right.

9 What is the focal length p of the parabola $x^2=12y$?

a. 3 ✓

b. 6

c. 12

d. 4

$4p=12\Rightarrow p=3$.

Section 4: Conic Sections (Ellipse & Hyperbola)

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10 What is the length of the major axis of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$?

a. 10 ✓

b. 5

c. 8

d. 20

$a^2 = 25 \Rightarrow a = 5$. Length of major axis $= 2a = 2(5) = 10$.

11 What is the eccentricity e of a conic section if it represents a hyperbola?

a. $e > 1$ ✓

b. $e = 1$

c. $0 < e < 1$

d. $e = 0$

By definition: Circle has $e = 0$, ellipse has $0 < e < 1$, parabola has $e = 1$, and hyperbola has $e > 1$.

12 What type of conic section is represented by $x^2 - 4y^2 - 2x + 16y - 19 = 0$?

a. **Hyperbola** ✓

b. Ellipse

c. Parabola

d. Circle

The coefficients of x^2 $(+1)$ and y^2 (-4) have opposite signs ($AC < 0$), which defines a hyperbola.

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